

From Classification to Generation: A Comparative Evaluation of VQA Architectures on VQA v2 and GQA

Davide Perniconi

Sapienza University of Rome

perniconi.1889270@studenti.uniroma1.it

Abstract

Visual Question Answering (VQA) requires answering natural language questions about images. We organize the VQA literature into three chronological architectural generations and empirically evaluate one representative model per generation: a CNN+LSTM baseline replicated from scratch, LXMERT, and BLIP, on VQA v2 and GQA. VQA v2 shows the expected progression (56.88% $\rightarrow$ 70.30% $\rightarrow$ 75.95%), with cross-modal attention providing the largest gain. On GQA, however, the ranking inverts: LXMERT (59.29%) outperforms BLIP (47.11%) because open-vocabulary generation without dataset-specific fine-tuning produces wrong answers. This suggests that dataset-specific training may contribute more to cross-dataset generalization than architectural generation alone.

1 Task Description

Visual Question Answering (VQA) requires a system to answer a natural language question about a given image. Formally, given an image I and a question Q , the system must produce an answer A [2]. The task was introduced as a benchmark for multimodal reasoning: to answer correctly, a model must jointly understand visual content and natural language, then align the two.

Answering visual questions involves at least three capabilities: (i) extracting visual features (objects, attributes, spatial relations); (ii) parsing question semantics, including compositional and comparative structures; (iii) cross-modal reasoning that connects linguistic queries to visual evidence.

1.1 Examples

Questions are typically grouped into three answer types:

- **Yes/No:** “Is the dog sitting on the couch?” $\rightarrow$ “yes” (object detection and spatial relation)
- **Number:** “How many people are in the image?” $\rightarrow$ “3” (instance detection and counting)
- **Other:** “What color is the car?” $\rightarrow$ “red” (attribute recognition and open-ended generation)

1.2 Real-world applications

VQA supports several downstream applications [10]. Assistive technology systems let visually impaired users ask questions about their surroundings, turning visual content into natural-language answers. In the medical domain, clinicians can query radiology scans (e.g., “Is there a fracture in this X-ray?”). Other documented use cases include educational tools that answer questions about diagrams or cultural-heritage artifacts, and video surveillance systems that answer natural-language queries about recorded footage.

2 Related Work

We organize the VQA literature into three broad architectural generations, reflecting its chronological evolution, and select one representative model per generation for our comparison.

2.1 CNN-RNN Fusion (2015–2018)

Antol et al. [2] introduced the VQA task alongside a baseline combining a VGGNet image encoder with an LSTM question encoder, reaching approximately 54% accuracy on the original VQA (v1) dataset. The approach relied on a simple fusion operation (element-wise product, addition, or concatenation) between the two modalities before

classification. This is the paradigm we replicate with our Vanilla VQA baseline: encode images with a CNN and questions with an RNN, then fuse and classify.

Anderson et al. [1] advanced this paradigm with Bottom-Up and Top-Down Attention, using Faster R-CNN region proposals [15] to identify salient image regions guided by the question. Their model reached approximately 65% on VQA v2. The key insight, that attention provides a learnable interface between vision and language, influenced all subsequent architectures.

2.2 Cross-Modal Transformers (2019–2021)

The Transformer [19] and BERT [3] changed how models handle cross-modal alignment. ViLBERT [12] introduced dual-stream encoders with cross-modal attention gates, reaching about 70.55% on VQA v2 test-dev. Rather than a single fusion step, attention is applied at multiple layers, letting the model build progressively refined image–question alignments.

LXMERT [18], our second-generation model, employs three Transformer encoders: one for object-level visual features, one for language, and one for cross-modal interaction. Like the Vanilla baseline, it answers by selecting from a fixed vocabulary via a classification head. With 36 pre-extracted Faster R-CNN regions per image, LXMERT learns alignments between specific objects and question words. On VQA v2 it reaches approximately 72%, with a separate checkpoint fine-tuned on GQA achieving 59.29% in our evaluation.

2.3 Generative Pretraining (2022–present)

Vision Transformers [4] showed that Transformers can match CNNs on visual tasks given enough data, enabling fully Transformer-based vision-language models. BLIP [11] combines a ViT image encoder with a BERT-based encoder-decoder, pretrained on large image-text corpora.

We use BLIP as our third-generation model. Unlike its predecessors, BLIP generates answers token-by-token rather than selecting from a fixed vocabulary. This open-vocabulary generation is both a strength (no vocabulary constraint) and a limitation: when applied zero-shot to GQA, where no GQA-specific checkpoint exists, it drops to 47.11%, well below LXMERT’s 59.29%. This gap is central to our comparative analysis.

3 Datasets and Benchmarks

We evaluate on two complementary benchmarks.

VQA v2 [5] is our primary benchmark with over 1.1 million questions about COCO images. Each question is answered by 10 annotators, enabling the soft accuracy metric that credits plausible alternatives. Critically, VQA v2 was designed to counter language bias: each question is paired with two similar images that yield different answers, forcing models to rely on visual content rather than linguistic shortcuts. The dataset provides test-dev and test-standard splits for official evaluation.

GQA [9] serves as a secondary benchmark focused on compositional reasoning. Its 22 million questions are generated from scene graphs, enabling control over linguistic diversity and reasoning complexity. We use the test-dev balanced split, which normalizes answer distributions to prevent trivial baselines from exploiting class imbalance. GQA allows us to test whether models trained on VQA v2 generalize to questions requiring structured reasoning about object attributes and relations.

Other benchmarks address complementary challenges not covered by our evaluation: OK-VQA [13] and A-OKVQA [16] require external or commonsense world knowledge beyond what is visible in the image; VizWiz [6] is sourced from photos and spoken questions recorded by blind users; TextVQA [17] requires reading and reasoning about text embedded in the image. We focus on VQA v2 and GQA as they remain among the most widely used benchmarks for comparing closed-vocabulary and generative VQA architectures.

4 Existing tools, libraries, papers with code

HuggingFace Transformers provides pretrained checkpoints for LXMERT and BLIP through a unified API, making Transformer-based models accessible without re-implementing complex architectures. The library handles tokenization, pre-processing, and model loading with a few lines of code.

NLTK is used in our baseline for word tokenization, paired with **GloVe** [14] (300-dimensional vectors from Common Crawl) to initialize the question embedding layer. Using pre-

trained embeddings accelerates LSTM convergence considerably on limited hardware.

Regarding papers with code: our Vanilla VQA baseline follows the architecture proposed by Antol et al. [2], whose official implementation is available in the [VQA-LSTM-CNN](#) repository; our answer normalization (contraction expansion, e.g. “aren’t”→“aren’t”, and number-word mapping, e.g. “one”→“1”) is adapted from the [official VQA evaluation code](#). LXMERT [18] and BLIP [11] also have official repositories ([airsplay/lxmert](#), [salesforce/BLIP](#)), but we use their pretrained checkpoints directly through HuggingFace rather than the original code. All code is included in the project repository: <https://github.com/the-summoning/nlp-project>.

5 State-of-the-art Evaluation

VQA v2 uses soft accuracy [2], which collects 10 annotator answers per question and awards partial credit proportional to annotator agreement: $\text{Acc}(a) = \min(\text{count}(a)/3, 1.0)$. If 7 of 10 annotators say “red” and 3 say “crimson”, predicting either earns full credit. This avoids penalising semantically equivalent answers and enables fine-grained breakdowns by question type: yes/no, number, and other.

GQA uses exact-match accuracy on the test-dev balanced split, with a single ground-truth answer per question after lowercasing. This is stricter than VQA v2’s soft metric: it penalises semantically correct answers phrased differently from the expected form (e.g., “a car” vs. “car”). The official breakdown separates binary (yes/no) from open questions; unlike VQA v2, GQA does not report a separate category for counting questions, which fall under “open”.

6 Comparative Evaluation

We compare three models spanning the three architectural generations: a Vanilla VQA baseline, LXMERT, and BLIP.

6.1 Experimental Setting

Vanilla VQA. We implemented this baseline from scratch following the architecture proposed by Antol et al. [2], with a few deliberate changes over the original design. We replace their VG-GNet image encoder with ResNet-101 [7] (pre-trained on ImageNet, final classification layer re-

moved), a stronger backbone unavailable at the time of the original paper; images are encoded offline and cached to disk as 2048-dimensional feature vectors, which also avoids loading the 44-million-parameter CNN during training, prohibitive on a single Kaggle T4 GPU. Questions are tokenized with NLTK and encoded by a 2-layer LSTM [8] whose embedding layer is initialized with GloVe [14] rather than learned from scratch: pretrained embeddings give the model better semantic representations from the start and speed up convergence on limited GPU time; the embeddings are then fine-tuned during training.

The fusion mechanism is a Hadamard (element-wise) product between L2-normalized image and question representations, each first projected to 1024 dimensions through an MLP with LayerNorm and Tanh. A two-layer classifier with LayerNorm, ReLU, and Dropout produces logits over a fixed answer vocabulary.

Two key design choices deserve mention. First, the answer vocabularies (3129 for VQA v2, 1842 for GQA) were chosen to match figures commonly used in VQA literature: large enough to cover most questions, but not so large that many answers would have only a handful of training examples each, too few to learn reliably given our limited GPU budget. Second, the model uses different loss functions per dataset: binary cross-entropy with soft labels for VQA v2 (where answers are inherently ambiguous) and standard cross-entropy for GQA (where each question has one ground-truth answer). Training uses Adam with learning rate 5×10^{-4} , batch size 256, and early stopping.

LXMERT. We evaluate two pretrained checkpoints from HuggingFace: [unc-nlp/lxmert-vqa-uncased](#) for VQA v2 and [unc-nlp/lxmert-gqa-uncased](#) for GQA. The model processes 36 pre-extracted Faster R-CNN region features per image (2048-dimensional) through three Transformer encoders. Like Vanilla VQA, LXMERT operates under closed-vocabulary classification: the output layer produces logits over a fixed answer set, and predictions are obtained via argmax. Inference is zero-shot: we use the provided classification heads without fine-tuning.

BLIP. We use the single available checkpoint [ybelkada/blip-vqa-base](#) from HuggingFace, loaded in `fp16` for memory efficiency. The model pairs a ViT-B/16 vision encoder with

Model	Y/N	Num	Other	Overall
<i>VQA v2 test-dev</i>				
Vanilla	75.23	35.70	46.03	56.88
LXMERT	86.76	52.95	60.17	70.30
BLIP	92.25	54.55	66.91	75.95
<i>VQA v2 test-standard</i>				
Vanilla	75.55	35.43	46.14	57.17
LXMERT	86.41	52.89	60.28	70.31
BLIP	92.32	54.39	66.81	76.03

Table 1: VQA v2 results (% soft accuracy, averaged over 10 annotators).

Model	Overall	Binary	Open
Vanilla	47.49	64.46	37.95
LXMERT	59.29	77.17	49.25
BLIP	47.11	65.66	36.69

Table 2: GQA test-dev balanced results (% exact-match accuracy).

a BERT-based encoder-decoder, generating answers via greedy decoding (max 10 tokens). Output is free-form text rather than a vocabulary index. Since no GQA-specific BLIP checkpoint exists, GQA evaluation is zero-shot on the VQA v2 checkpoint.

6.2 Evaluation Protocol

For VQA v2, predictions on test-dev and test-standard are submitted to the Eval.AI server¹, which computes accuracy with per-type breakdowns. For GQA, the official competition is closed, so we evaluate directly on the test-dev balanced split by comparing predictions to ground-truth answers, reporting overall, binary (yes/no), and open accuracy.

6.3 Results

VQA v2 (Table 1) shows the expected generational progression: 56.88% (Vanilla) $\rightarrow$ 70.30% (LXMERT, +13.42) $\rightarrow$ 75.95% (BLIP, +5.65). The largest gains appear between the Vanilla baseline and LXMERT, suggesting that cross-modal attention contributes more than the shift from classification to generation. All three models follow the same difficulty ranking: yes/no questions are easiest, counting is hardest.

GQA (Table 2) tells a different story. LXMERT leads at 59.29%, while BLIP drops to 47.11%, below even the Vanilla baseline (47.49%). On the

binary subset, LXMERT holds a 12-point advantage over BLIP (77.17% vs. 65.66%), and the gap widens on open questions (49.25% vs. 36.69%). Note that GQA’s “open” category, unlike VQA v2’s “other” column in Table 1, also includes counting questions, so the two are not directly comparable.

6.4 Discussion

The VQA v2 results are consistent with the architectural narrative. The Vanilla baseline’s Hadamard fusion captures only first-order interactions between modalities; LXMERT’s multi-layer cross-attention builds richer, context-dependent alignments; BLIP’s generative pretraining adds a further boost, particularly on open-ended questions where its vocabulary is not artificially restricted. Our reproduced BLIP accuracy (75.95%) is below the 78.25% reported in the original paper [11], plausibly due to two protocol differences rather than a flaw in our evaluation: the original paper fine-tunes and evaluates at 480×480 resolution, while our pipeline operates at 224×224 , giving the model less visual detail; and the official BLIP codebase evaluates VQA by *ranking* a fixed list of candidate answers, whereas our setup uses free-form greedy *generation*, which is not constrained to match the exact wording of any ground-truth answer and is therefore more exposed to the exact-match penalty on semantically correct but differently phrased answers.

The GQA results demand a more nuanced reading. Unlike the modest VQA v2 gap discussed above, BLIP’s much larger GQA shortfall cannot be attributed to evaluation protocol (rank vs. generation) or resolution alone; it stems primarily from zero-shot domain transfer, detailed next. LXMERT’s advantage is not purely architectural, since it benefits from a checkpoint specifically fine-tuned on GQA. BLIP, lacking any GQA adaptation, must answer GQA questions using only its VQA v2 pretraining: on questions requiring compositional reasoning (e.g., “What color is the car to the left of the tree?”), BLIP’s free-form generation often misses the relational constraint, whereas LXMERT’s fixed-vocabulary classification is constrained to answers the dataset expects. GQA’s exact match further penalises open-vocabulary models for syntactic variation; BLIP is additionally disadvantaged by the absence of answer normalisation on its free-form output, unlike

¹Link to the VQA v2 challenge: <https://eval.ai/web/challenges/challenge-page/830/overview>

the fixed, canonicalised vocabularies of Vanilla and LXMERT. The expected gain from normalisation is small, however, and does not alter the ranking.

This reversal, with BLIP dominating VQA v2 but underperforming on GQA, illustrates a tension between generality and specialization. Generative models offer flexibility (answers beyond a fixed vocabulary) and strong in-domain performance, but their advantage shrinks when the evaluation distribution differs from pretraining. Closed-vocabulary classifiers trained on task-specific data remain competitive, particularly on benchmarks with controlled answer spaces.

A further limitation applies to all three models. Vanilla VQA and LXMERT are also capped by a fixed answer vocabulary: even if they reason correctly about the image, they can only output the closest available answer, so a wrong prediction does not necessarily mean an absence of reasoning. Still, the pattern is consistent with the well-known language-prior problem in VQA, where models partly guess from question wording rather than the image. BLIP is not restricted to a fixed list of answers, but its sharp drop on GQA suggests it relies on something similar: its answers still seem to depend on the patterns it picked up from VQA v2, since a model that truly reasoned about the image and question should not collapse so badly on a new dataset.

Counting remains a hard ceiling for all models. Even BLIP, the strongest system, reaches only 54.55% on VQA v2 counting questions, compared to 92.25% on yes/no. This gap reflects a persistent limitation of current architectures: detecting objects is necessary but not sufficient for counting them.

7 Conclusions

We compared three VQA architectures across two benchmarks, covering a decade of progress from CNN+LSTM fusion to generative pretrained models. On VQA v2 the progression is unambiguous: 56.88% $\rightarrow$ 70.30% $\rightarrow$ 75.95%. On GQA the ranking reverses, with LXMERT (59.29%) outperforming BLIP (47.11%) by a wide margin.

Three conclusions emerge. First, cross-modal attention (the transition from CNN-RNN fusion to LXMERT) accounts for the largest single performance jump on VQA v2, more than the subsequent shift to generation, suggesting that fine-

grained alignment between image regions and question tokens is the central challenge in VQA. Second, generative pretraining does not automatically transfer across datasets: without GQA-specific fine-tuning, BLIP’s open-vocabulary answers no longer match GQA’s expected answer distribution, and BLIP falls below even the Vanilla baseline. Third, at least Vanilla VQA and LXMERT appear to work mainly by picking the answer that matched similar patterns during training, rather than truly reasoning about the image and question, and BLIP’s own collapse on GQA suggests it relies on the same kind of pattern matching to some extent.

8 Use of GenAI and LLM tools

Generative AI tools were used in two distinct, disclosed roles. ChatGPT (OpenAI) assisted while writing the three model notebooks (`vanilla-vqa.ipynb`, `lxmert.ipynb`, `blip.ipynb`): drafting PyTorch/HuggingFace data-loading, training, and evaluation code, and debugging runtime errors. Claude Code (Anthropic) assisted with this report: drafting and revising prose for clarity and academic style, formatting results into LaTeX tables, fixing LaTeX compilation issues, and cross-checking cited accuracy figures and bibliographic entries against the original papers. All experimental design choices, the framing adopted to organize the related work, and the interpretation of results are the author’s own; all AI-assisted code and text were reviewed by the author, who is responsible for the final content.

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